# Introduction to Artificial Intelligence

Estimated time: 45 minutes • Beginner Level

## What You'll Learn

- Understand what Artificial Intelligence is and how it works
- Identify the key types of AI: narrow AI, general AI, and superintelligence
- Explore real-world applications of AI in business and everyday life
- Recognize the difference between AI, Machine Learning, and Deep Learning

## 1. What is Artificial Intelligence?

Artificial Intelligence (AI) refers to the simulation of human intelligence processes by machines, especially computer systems. These processes include learning, reasoning, and self-correction. AI was first coined as a term in 1956 by John McCarthy at the Dartmouth Conference.

### *Key Definition*

**AI** is the ability of a digital computer or computer-controlled robot to perform tasks commonly associated with intelligent beings — such as the ability to reason, discover meaning, generalize, or learn from past experience.

## 2. Types of Artificial Intelligence

| Type | Description | Example |
|---|---|---|
| Narrow AI | Designed for a specific task | Siri, Google Translate |
| General AI | Performs any intellectual task | Still theoretical |
| Super AI | Surpasses human intelligence | Hypothetical future |

## 3. AI vs Machine Learning vs Deep Learning

People often confuse these three terms. Here's how they relate to each other:

- **Artificial Intelligence** — The broadest concept. Any technique enabling machines to mimic human behaviour.

- **Machine Learning** — A subset of AI. Algorithms that learn from data without being explicitly programmed.
- **Deep Learning** — A subset of ML. Uses neural networks with many layers to learn from large datasets.

### *A Simple Analogy*

Think of AI as the universe, Machine Learning as the solar system within it, and Deep Learning as planet Earth within that solar system.

## 4. Real-World Applications of AI

**Healthcare:** Diagnosing diseases, drug discovery, personalised treatment plans

**Finance:** Fraud detection, algorithmic trading, credit scoring

**Marketing:** Personalised recommendations, customer segmentation, chatbots

**Education:** Adaptive learning platforms, automated grading, tutoring systems

**Transport:** Self-driving vehicles, route optimisation, traffic prediction

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## Key Takeaway

AI is not magic — it is mathematics and data working together. Understanding its foundations empowers you to apply it strategically in your work and career. In the next module, we will dive deeper into how machine learning models are trained and what they can do for your business.